Quantum Circuit-Based Adaptation for Credit Risk Analysis

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Noisy and Intermediate-Scale Quantum, or NISQ, processors are sensitive to noise, prone to quantum decoherence, and are not yet capable of continuous quantum error correction for fault-tolerant quantum computation. Hence, quantum algorithms designed in the pre-fault-tolerant era cannot neglect the noisy nature of the hardware, and investigating the relationship between quantum hardware performance and the output of quantum algorithms is essential. In this work, we experimentally study how hardware-aware variational quantum circuits on a superconducting quantum processing unit can model distributions relevant to specific use-case applications for Credit Risk Analysis, e.g., standard Gaussian distributions for latent factor loading in the Gaussian Conditional-Independence model. We use a transpilation technique tailored to the specific quantum hardware topology, which minimizes gate depth and connectivity violations, and we calibrate the gate rotations of the circuit to achieve an optimized output from quantum algorithms. Our results demonstrate the viability of quantum adaptation on a small-scale, proof-of-concept model inspired by financial applications and offer a good starting point for understanding the practical use of NISQ devices. [1]

I. Introduction

Quantum computing has been promised as a reliable solution for use cases that are inefficiently addressed by standard classical computing. Among them, financial applications have attracted considerable attention lately. In the Credit Risk Analysis (CRA) domain, quantum strategies [1]–[3] for computing quantities related to the estimation of Economic Capital, such as the Value at Risk (VaR) and the Conditional Value at Risk (CVaR), rely on Quantum Amplitude Estimation (QAE). This technique promises a theoretical quadratic speed-up with respect to its classical counterpart, the Monte Carlo simulation approach [4]. However, the reduced number of addressable qubits and the relatively high level of gate errors and decoherence in state-of-the-art Quantum Processing Units (QPUs) represent a technological challenge to overcome for the demonstration of quantum algorithms, at least as proof-of-concepts [5]. Nevertheless, great advances in hardware performance enable us to explore several use cases with remarkable results already in the Noisy and Intermediate-Scale Quantum (NISQ) era [5]. Especially relevant in this framework are superconducting QPUs, which have attracted noticeable interest worldwide for their remarkable circuit analogies with CMOS-based classical processors, sharing similar historical bottlenecks, advantages, limitations, perspectives, and compatibilities with state-of-the-art classical computing techniques [6]. Remarkably, superconducting technology provides extensive engineering freedom, which allows for an enriching day by day pool of applications in the quantum technologies realm [7], [8] and a strong variety of hardware platforms for quantum computing, each designed to tackle specific limitations of current quantum circuit designs [9]. Superconducting QPUs (sQPUs) leverage the use of Josephson junctions to artificially engineer macroscopic artificial atoms in a way that allows easy preparation, manipulation, and readout of quantum states with an increasing level of complexity [10]–[12], and eventually exploit the fundamental computational resources of quantum computing: superposition and entanglement. Most importantly, the practical interface between sQPUs and classical computing platforms, e.g., cloud computing and High-Performance Computing (HPC), has provoked a strong speed-up and quantum awareness in the field of quantum algorithms [13], [14]. Although pushing coherence times well above current limits and inventing smart ways to overcome gate and decoherence errors [15]–[17] is probably the most sound solution to reach the fault-tolerant quantum (FTQ) regime [18], it is still worth

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systematically and coherently investigating the possible outcomes achievable in NISQ devices.

In this work, we exploit a transmon-based sQPU, the core of the Italian Superconducting Quantum Computing Center *Partenope*, conceived to experimentally address quantum algorithms directly at the hardware (or pulse) level. We report here the first experimental investigation and implementation of one sub-circuit in the QAE-based Credit Risk Analysis algorithm on this quantum machine. In this quantum procedure, the first part of the Grover operator's state preparation is dedicated to loading an *uncertainty model* that represents the probability of default of the counterparties under analysis. To perform preliminary experiments on quantum hardware, we decided to focus on this component since it remains unchanged when estimating both the VaR and the CVaR through repeated applications of the Grover operator. Moreover, the construction of the uncertainty model requires the loading of standard normal distributions $\mathcal{N}(0,1)$, with a mean equal to 0 and a standard deviation of 1, a task that is also reusable for other quantum finance use cases, such as option pricing [19]–[22]. Concerning the execution of the *uncertainty model*, the *transpilation*, i. e. the process of converting a quantum circuit from one representation to another (often to optimize it for a specific quantum hardware platform) plays a key role, especially in the NISQ era and in QPUs characterized by limited connectivity (as the sQPU analyzed in this work). Akin to compilation in classical computing, transpilation adapts quantum circuits for the unique characteristics and limitations of different quantum computers instead of transforming code for different processors. Although in the long term quantum computing should be hardware agnostic, it is still not the case for NISQ devices. This work, in fact, aims to bridge a fundamental gap between quantum algorithm specialists and quantum hardware engineers, discussing in detail at the machine level the role of (i) hardware circuit parameters, (ii) qubit connectivity, (iii) hardware performance, (iv) electronics implementation of quantum logic gates on the actual output of a quantum algorithm, and the transpilation process that allows optimizing the outcome of a quantum circuit.

The paper is organized as follows. Sec. II defines how the uncertainty model is mapped to a quantum circuit. We present the scalable, parameterized circuit for a single-counterparty model, whose probability of default is influenced by a single risk-factor, and describe how each component of the circuit was developed. Sec. III details the hardware we used for the executions — qubit topology and gate fidelities — and shows the transpiled circuit that reflects the device's coupling map, which differs from the idealized layout. This allows us to detail the connection between the simulated transpilation of the quantum circuits and the effective transpilation required by the machine to achieve optimal results. Sec. IV presents results and discussion, comparing the quantum implementation with a classical baseline for the same model.

II. Methods

A. Context of Credit Risk Analysis and Gaussian Conditional Independence Model

CRA quantifies the likelihood and impact of borrower default at both single-exposure and portfolio levels. The core ingredients are the probability of default (PD) and the loss given default (LGD), which combine to determine expected and tail losses over a horizon. Since debtors are not independent, portfolio models introduce latent risk factors to capture dependence (e.g., one-factor Vasicek or multi-factor structures, often paired with copulas for joint tails). The accurate simulation of correlated default scenarios and loss distributions is fundamental for practical tasks, e.g., pricing and capital allocation, as well as Basel stress testing. The principal computational bottleneck is the generation of high-fidelity samples from these correlated models and the evaluation of portfolio losses across many scenarios.

These requirements naturally favor latent-factor formulations that balance realism with tractability. The Gaussian conditional-independence (GCI) *uncertainty model* [2] provides an analytically convenient link from shared risk factors to debtor-level default probabilities and is widely used as a classical baseline [24], [25]. Classically, GCI posits that the debtor-level conditional default probabilities are conditioned on one (or several) latent normal risk factor(s) z as:

$$PD_k(z) = \Phi\left(\frac{\Phi^{-1}(p_k^0) - \sqrt{\rho_k}z}{\sqrt{1-\rho_k}}\right), \tag{1}$$

where Φ is the standard normal Cumulative Distribution Function (CDF), p_k^0 and ρ_k are, respectively, the baseline default level and the correlation of the k-th debtor (asset).

For the quantum mapping of this uncertainty model, we define the conditional default probability $PD_k(z)$ as the probability of measuring the state $|1\rangle$ on a target qubit,

$$PD_k(z) \equiv P_1 = \sin^2\big(\alpha_k z + \beta_k\big). \tag{2}$$

Here, the parameters α_k and β_k are obtained by representing the classical conditional default probability function in a linear form within the latent factor z, such that the corresponding quantum rotation angle depends affinely on z with a slope α_k and an offset β_k. In this way, conditional default mechanisms can be implemented by parameterizing single-qubit rotations of the type

$$R_y\big(2(\alpha_k z + \beta_k)\big), \tag{3}$$

so that the amplitude of $|1\rangle$ encodes the desired conditional default probability.

In the circuit, the latent factor z is represented by an n-qubit register, whose computational basis states encode a discrete set of latent factor values. Absorbing these discrete values into effective linear parameters with slope $\tilde{\alpha}_k$ and offset $\tilde{\beta}_k$, the gate implemented on the target qubit can be written compactly as

$$R_y\big(2(\tilde{\alpha}_k\, \hat{z}_{\text{code}} + \tilde{\beta}_k)\big), \tag{4}$$

where $\hat{z}_{\text{code}}$ denotes the integer value encoded by the n-qubit z-register and controls the rotation angle, thereby encoding

the conditional default probabilities $PD_k(z)$ across the encoded configurations (see Supplementary Material for further details).

In this work, we focused on a scenario involving one asset and one latent factor employing three qubits: q_0 and q_1 form the z-register ($n = 2$) that prepares a discrete normal distribution over the latent factor z, while q_2 is the asset qubit that encodes the conditional default probability.

Loading of the latent factor uses two single-qubit $R_y(\cdot)$ rotations followed by a $\mathrm{CNOT}_{0,1}$ entangler, shaping the two-qubit amplitudes to approximate a Gaussian over the (q_0, q_1) register (the green square). The state of the asset (default) qubit q_2 can be prepared via controlled $R_y(\cdot)$ rotations conditioned on q_0 and q_1, such that the effective rotation angle, and consequently the probability of measuring $|1\rangle$ on q_2, depends on the encoded value of the normal factor. This realizes the conditional map $PD(z) \approx \sin^2(\alpha z+\beta)$, linking the risk factor to the asset's default probability within a shallow, hardware-efficient circuit.

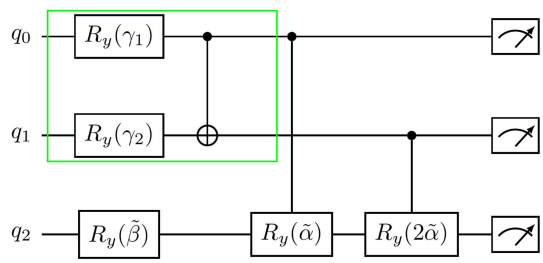


Fig. 1: Quantum circuit for the target GCI model with one asset and one risk factor. The green box underlines the Gaussian loading sub-circuit.

B. Variational circuit for a normal distribution

A crucial role in the CRA and GCI models is played by the preparation of a discrete normal distribution over the latent factor. In fact, the GCI quantum circuit in Fig. 1 includes a Gaussian loading sub-circuit on a register of 2 qubits. The generalization to n qubits, spanning 2^n computational basis states, yields the requirement to prepare a quantum state $\psi(\boldsymbol{\theta})$,

$$|\psi(\boldsymbol{\theta})\rangle \;=\; \sum_{b=0}^{2^n-1} \sqrt{p_b(\boldsymbol{\theta})}\,|b\rangle, \tag{5}$$

so that measuring in the computational basis returns the bitstring b with a probability of $p_b(\boldsymbol{\theta})$.

We first adopt a distribution-agnostic strategy: select a hardware-efficient ansatz and train its parameters $\boldsymbol{\theta}$ so that the model distribution $p(\boldsymbol{\theta})$ matches a prescribed histogram

$$p^\star = \{p_b^\star\}_{b=0}^{2^n-1}. \tag{6}$$

Given a discrepancy functional $D\big(p(\boldsymbol{\theta}) \,\|\, p^\star\big)$, gradients are estimated via the parameter–shift rule, and $\boldsymbol{\theta}$ is updated with a stochastic optimizer such as Adam [26], [27]. This recipe—compact ansatz, target histogram, shift-rule gradients, and Adam updates—supports arbitrary discrete distributions over the register.

As concrete instances, we synthesize discrete Gaussians on two and three qubits.

Two qubits. On the register (q_0, q_1), we use the hardware-efficient ansatz: single-qubit $R_y(\cdot)$ rotations on each qubit followed by one entangler $\mathrm{CNOT}_{0,1}$ (Fig 2). Let $\boldsymbol{\theta}$ collect all rotation angles; the Born probabilities are

$$p_b(\boldsymbol{\theta}) = \big|\langle b \mid \psi(\boldsymbol{\theta})\rangle\big|^2, \qquad b \in \{0,1,2,3\}. \tag{7}$$

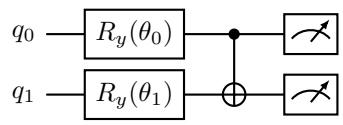


Fig. 2: Two-qubit quantum circuit for the Gaussian distribution loading.

Three qubits. We extend the circuit to (q_0, q_1, q_2) by adding an $R_y(\theta_2)$ on q_2 and a second entangler $\mathrm{CNOT}_{0,2}$ (control q_0) (Fig 3). The output probabilities are

$$p_b(\boldsymbol{\theta}) = \big|\langle b \mid \psi(\boldsymbol{\theta})\rangle\big|^2, \qquad b \in \{0,1,\ldots,7\}. \tag{8}$$

Target and training. For either $n \in \{2,3\}$ qubits, we map

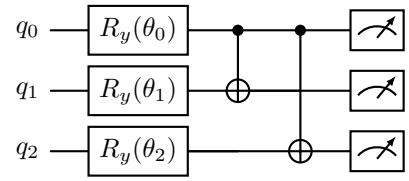


Fig. 3: Three-qubit quantum circuit for the Gaussian distribution loading.

each basis index b to a grid point $z(b) \in [-z_{\max}, z_{\max}]$ via an affine transform and define the target histogram of $\mathcal{N}(\mu, \sigma^2)$ as

$$p_b^\star \propto \exp\Big(-\frac{(z(b)-\mu)^2}{2\sigma^2}\Big), \qquad b = 0,\ldots,2^n-1. \tag{9}$$

Parameters are learned by minimizing the distributional loss

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{b=0}^{2^n-1} \Big(p_b(\boldsymbol{\theta}) - p_b^\star\Big)^2, \tag{10}$$

using parameter-shift gradients and Adam updates.

After convergence, the measured histograms for the two- and three-qubit circuits closely match the target normal specified by (μ, σ), yielding shallow, hardware-efficient loaders suitable for subsequent calibration and hyperparameter tuning on the sQPU.

C. Hardware-Level normal distribution preparation: two and three Qubits

Starting with a two-qubit system, we study how the gates affect the probability amplitudes of the basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. The single–qubit rotation

$$R_Y(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \tag{11}$$

acts on each qubit. From the initial state $|00\rangle$, whose amplitude vector is $[1,0,0,0]^\top$, we obtain

$$\begin{matrix}|00\rangle\\|01\rangle\\|10\rangle\\|11\rangle\end{matrix}\quad\begin{bmatrix}1\\0\\0\\0\end{bmatrix}\xrightarrow{R_Y(\theta_0)\otimes R_Y(\theta_1)}\begin{bmatrix}\cos(\theta_0/2)\cos(\theta_1/2)\\\cos(\theta_0/2)\sin(\theta_1/2)\\\sin(\theta_0/2)\cos(\theta_1/2)\\\sin(\theta_0/2)\sin(\theta_1/2)\end{bmatrix}.$$

Applying a CNOT with control q_0 and target q_1 yields

$$\begin{bmatrix}\cos(\theta_0/2)\cos(\theta_1/2)\\\cos(\theta_0/2)\sin(\theta_1/2)\\\sin(\theta_0/2)\sin(\theta_1/2)\\\sin(\theta_0/2)\cos(\theta_1/2)\end{bmatrix}, \tag{12}$$

i.e., the amplitudes of $|10\rangle$ and $|11\rangle$ are swapped. Consequently, the output probabilities depend on the choices of θ_0 and θ_1.

For a symmetric, bell-shaped target (e.g., a discrete Gaussian), we require

$$P_{|00\rangle}=P_{|11\rangle},\qquad P_{|01\rangle}=P_{|10\rangle},\qquad P_{|00\rangle,|11\rangle}<P_{|01\rangle,|10\rangle}.$$

From Eq. (12), these symmetries are satisfied when

$$\theta_0=\pm\frac{(2n_0+1)\pi}{2},\theta_1=2\pi n_1\pm\frac{(2n_0+1)\pi}{2},n_0,n_1\in\mathbb{N}, \tag{13}$$

and the "central-mass" condition $P_{|00\rangle,|11\rangle}<P_{|01\rangle,|10\rangle}$ further imposes

$$\cos(\theta_1/2)<\sin(\theta_1/2)\iff\pi/2<\theta_1<3\pi/2\pmod{2\pi}. \tag{14}$$

By taking the software-optimized loader as our baseline, we can then fix θ_0 as the reference angle and tune θ_1 to match the characteristics of the quantum hardware at hand. The output profile is uniform when $\theta_1=\{\pi/2,\,5\pi/2,\ldots\}$, while to obtain a Gaussian-like shape, one can choose θ_1 so that $\cos(\theta_1/2)<\sin(\theta_1/2)$, which yields $P_{|00\rangle,|11\rangle}<P_{|01\rangle,|10\rangle}$.

Extension to Three Qubits. We now scale to a three-qubit register with basis $\{|000\rangle,|001\rangle,|010\rangle,|011\rangle,|100\rangle,|101\rangle,|110\rangle,|111\rangle\}$. Starting from the two-qubit output (with $\theta_0=\pi/2$) and adding an $R_Y(\theta_2)$ on q_2, then a second entangler $\mathrm{CNOT}_{0,2}$ (control q_0, target q_2), the amplitude trajectory reads

$$\frac{1}{\sqrt{2}}\begin{bmatrix}\cos(\theta_1/2)\\0\\\sin(\theta_1/2)\\0\\\sin(\theta_1/2)\\0\\\cos(\theta_1/2)\\0\end{bmatrix}\xrightarrow{R_Y(\theta_2)}\frac{1}{\sqrt{2}}\begin{bmatrix}\cos(\theta_1/2)\cos(\theta_2/2)\\\cos(\theta_1/2)\sin(\theta_2/2)\\\sin(\theta_1/2)\cos(\theta_2/2)\\\sin(\theta_1/2)\sin(\theta_2/2)\\\sin(\theta_1/2)\cos(\theta_2/2)\\\sin(\theta_1/2)\sin(\theta_2/2)\\\cos(\theta_1/2)\cos(\theta_2/2)\\\cos(\theta_1/2)\sin(\theta_2/2)\end{bmatrix}$$

$$\xrightarrow{\mathrm{CNOT}_{0,2}}\frac{1}{\sqrt{2}}\begin{bmatrix}\cos(\theta_1/2)\cos(\theta_2/2)\\\cos(\theta_1/2)\sin(\theta_2/2)\\\sin(\theta_1/2)\cos(\theta_2/2)\\\sin(\theta_1/2)\sin(\theta_2/2)\\\sin(\theta_1/2)\sin(\theta_2/2)\\\sin(\theta_1/2)\cos(\theta_2/2)\\\cos(\theta_1/2)\sin(\theta_2/2)\\\cos(\theta_1/2)\cos(\theta_2/2)\end{bmatrix}.$$

The combined action of $\mathrm{CNOT}_{0,1}$ and $\mathrm{CNOT}_{0,2}$ (control q_0 in both cases) swaps the pairs $|100\rangle\leftrightarrow|101\rangle$ and $|110\rangle\leftrightarrow|111\rangle$. Under the same symmetry conditions on θ_0 and θ_1 as above, a symmetric three-qubit profile requires

$$P_{|000\rangle}=P_{|111\rangle},\;\;P_{|001\rangle}=P_{|110\rangle},\;\;P_{|010\rangle}=P_{|101\rangle},$$

$$P_{|011\rangle}=P_{|100\rangle},$$

which is obtained when

$$\theta_2=2\pi n_2\pm\theta_1,\qquad n_2\in\mathbb{N}. \tag{15}$$

To enforce a central tendency (highest mass near $|000\rangle$ and $|111\rangle$) with strictly decreasing "rings",

$$P_{|000\rangle,|111\rangle}\;<\;P_{|001\rangle,|110\rangle}\;<\;P_{|010\rangle,|101\rangle}\;<\;P_{|011\rangle,|100\rangle},$$

one further selects θ_2 so that $\cos(\theta_2/2)<\sin(\theta_2/2)$, i.e.

$$\pi/2<\theta_2<5\pi/2\pmod{4\pi}. \tag{16}$$

These analytic angle relations provide hardware-level guidance for shaping symmetric, bell-like histograms with shallow entangling patterns on today's sQPUs.

In Sec. IV, we examine how sQPU connectivity and circuit choices (e.g., entangler placement and single-qubit rotation settings) affect the quality of the prepared distribution.

III. Hardware description

The superconducting Quantum Processing Unit (sQPU) used in this work is a Contralto-D from Quantware (Fig. 4), composed of 17 operational symmetric flux-tunable transmons, connected in a rectangular 2D matrix through fixed high-frequency bus resonators.

Qubits are designed to fall into three frequency bandwidths: high-frequency qubits have frequencies of the order of 6 GHz (in red), medium frequency qubits have frequencies of the order of 5 GHz (in cyan), and low frequency qubits lie in a frequency range of the order of 4 GHz (in green). All the qubits are equipped with a dedicated superconducting readout resonator, equally distributed over 4 readout feedlines, thereby guaranteeing multiplexed readout. They also use dedicated drive and flux lines to implement single-qubit X-Y gates and Z-gates, as well as the native two-qubit gate of the sQPU, the Conditional-Z (CZ) gate, which we implement by using Sudden-Net-Zero (SNZ) flux pulses on the highest-frequency qubit in a pair [28]. Flux lines also allow for parking the qubits at specific working points by using static magnetic flux. In this work, we have focused on qubits C4, D3, and A6 (low-, medium-, and high-frequency qubits, respectively), which have been operated at their flux sweet-spots. The remaining qubits of the matrix have been operated at their anti-sweet spot, i. e., nominally at zero frequency, to limit the effect of frequency crowding and qubit crosstalk.

The sQPU is installed at the coldest stage of a Bluefors XLD1000SL, equipped with cryogenic attenuated drive and

	T_1 (μs)	T_2^* (μs)	T_{2E} (μs)	F_{avg} (%)	F_{RO} (%)	F_{CZ} (%)
A6	43 ± 1	37 ± 3	40 ± 1	99.855 ± 0.005	0.95 ± 0.01	
C4	38 ± 1	20 ± 1	42 ± 4	99.924 ± 0.002	0.95 ± 0.01	
D3	32 ± 1	20 ± 6	37 ± 6	99.91 ± 0.01	0.94 ± 0.01	
A6-D3						99.7 ± 0.2
D3-C4						99.05 ± 0.07

TABLE I: Contralto-D QPU specifications for A6, D3, C4 qubits: coherence times (T_1 - relaxation time, T_2^* - Ramsey coherence time, T_{2E} - Hahn-Echo coherence time, with errors provided by 100 repeated experiments), average single-qubit gate (F_{avg}) and readout (F_{RO}) fidelities, and two-qubit CZ interleaved average gate fidelity F_{CZ} for the pairs. In green low frequency band qubits, in blue mid frequency band qubits and in red high frequency band qubits.

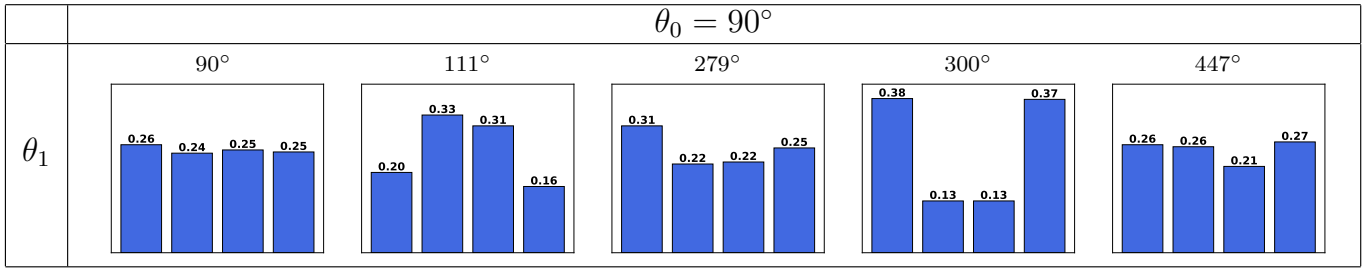


TABLE II: Two-qubit circuit output on D3-C4 register as a function of θ_1 and fixed $\theta_0 = 90°$. For $\theta_1 \in \{90°, 270°\}$ probability amplitudes feature a gaussian-like shape, while for $\theta_1 \in \{270°, 450°\}$ the concavity is inverted.

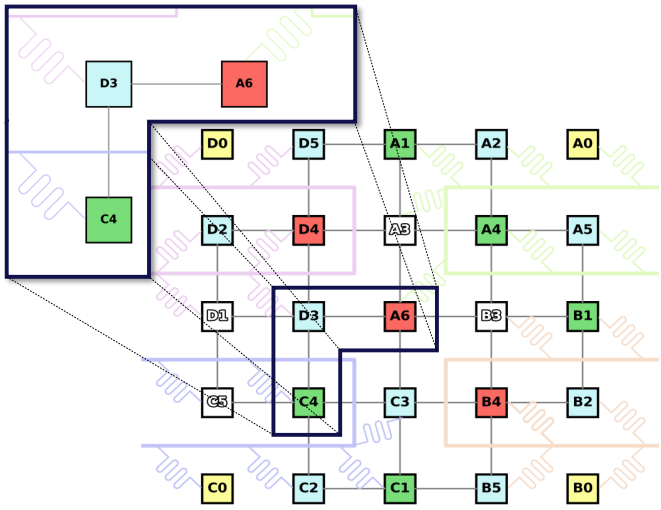


Fig. 4: Schematics of the processor, with focus on three qubit register A6, D3, C4. The processor has 4 feedlines for multiplexed readout: feedline A (light green), coupled to 6 qubits, and feedlines B (pink), C (indigo), and D (purple), each coupled to 5 qubits. In green: low frequency qubits; in blue: mid frequency qubits; in red: high frequency qubits; in yellow: isolated qubits. White color identifies qubits that are not operational.

read-in lines, superconducting cryogenic coaxial flux lines, and read-out lines with High-Electron Mobility Transistors (HEMT) amplifiers, anchored at the 4K plate of the cryostat, and a double junction isolator at the MXC. Additional detail on the cryogenic experimental setup is reported in the Supplementary Material of Ref. [29]. At room temperature, the sQPU interfaces with a host PC through LAN connections via a Qblox cluster, equipped with FPGA-based modules for control, read-out, and flux pulsing of qubits. Static flux-pulses for qubits parking are delivered by low-noise DC electronics, the SPI rack, and combined with flux-pulsed signals from the cluster using bias-tees at room temperature. The Qblox electronics can be controlled by a Python-based framework, *Quantify* [30], which allows us to design experimental schedules to run quantum circuits of varying complexity, control every physical quantity of X, Y, Z, and decompose the required two-qubit CNOT gate in terms of the native CZ gate directly at the pulse-level. Specifically, in our sQPU, X and Y gates are implemented using Derivative Reduction Adiabatic Gate (DRAG) microwave pulses of 60 ns: rotation angles are set by changing the amplitude of such pulses, while the rotation axis is selected by changing the phase of the microwave signals. Microwave pulses are also used to excite the dedicated superconducting resonators, and readout is performed by digitizing the output signal through the readout modules. Meanwhile, Z single-qubit gates are virtually implemented by phase updates of the control signal [31], [32]. All other single-qubit rotations are generally decomposed in terms of the X, Y, and Z rotations. Additional details can be found in [10].

In Tab. I, we report on the coherence times, gate, and readout fidelities of the investigated qubits, acquired right before the experiments to have available benchmarking coherence and fidelity parameters of the NISQ machine. This comes in handy when we want to investigate any relation between a quantum algorithm output and the noise of the NISQ processor [29]. Additional details on calibration procedures and the gate and readout fidelities, as well as the assignment

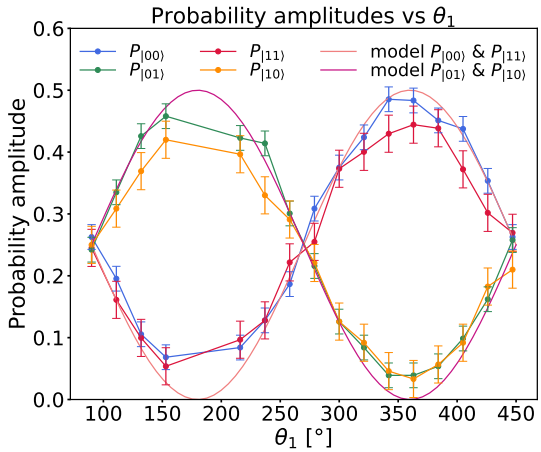


Fig. 5: Comparison between probability amplitudes of $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ states and theoretical model as a function of $\theta_1 \in \{90°, 450°\}$ and fixed $\theta_0 = 90°$.

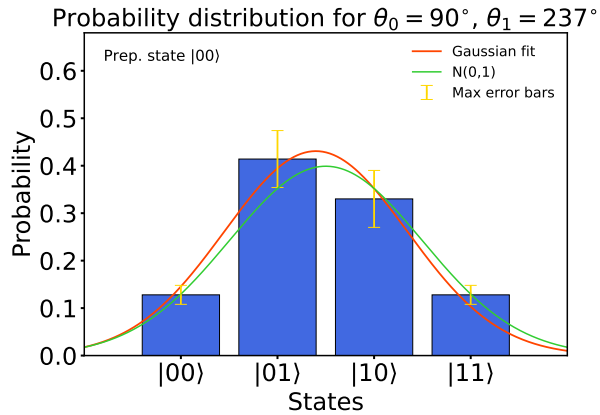
calibration matrices for the experiments reported in this work, are collected in the Supplementary Material.

IV. Results and Discussion

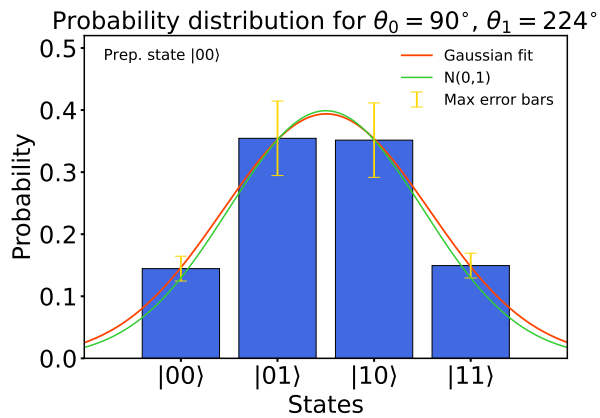
A. Two-qubit normal distribution encoding and loading

The first experimental investigation involved the D3-C4 pair, and the goal was to identify the best θ_0, θ_1 angles combination to achieve a two-qubit standard normal distribution. By preparing the two-qubit system in the $|00\rangle$ state, we first applied an $R_y(\cdot)$ gate on D3 with fixed $\theta_0 = 90°$ and $\theta_1 = 111°$, obtained through the target and training procedure and the Adam optimization discussed in Sec. II-B. The experimental result on the quantum machine is reported in Tab. II. With the goal of experimentally assessing at the hardware level the effect of the rotation angle on the probability distribution shape, we swept the angle θ_1 for the $R_y(\cdot)$ gate on C4 in the range $[90°, 450°]$ with a step of $21°$. To provide a comparison with the initial experiment, in Tab. II we collect some of the acquired output distributions to show that they exhibit a periodic trend: for $90° < \theta_1 < 270°$, the probability amplitude distributions show a positive concave shape; for $270° < \theta_1 < 450°$, the concavity inverts, while for θ_1 around $90°$, $270°$, and $450°$, the distribution becomes uniform. The experimental probability amplitudes for the two-qubit basis states as a function of all the angles θ_1 are also reported in Fig. 5, showing a reasonable agreement with the theoretical expectation. This first screening allowed us to identify a promising region with a positive concave shape and probabilities compatible with the simulated output. We repeated the experiment by thickening the step to $1°$ between these angles to find the optimal θ_1, which resulted in $\theta_1 = 237°$ (Fig. 6 (a)). Similar experiments were also performed by preparing $|11\rangle$ as the initial state, which can be found in the Supplementary Material.

As one can notice, the quantum circuit output can reveal fluctuations that may make the output distribution slightly asymmetric. This can be accounted for State Preparation And Measurement (SPAM) errors in an NISQ sQPU. Therefore,



(a)



(b)

Fig. 6: Two-qubit Gaussian distribution at the optimal θ_0 and θ_1 angle and corresponding Gaussian fit for two different pairs. In (a), experimental results for pair D3-C4. In (b), experimental results for pair D3-A6. In green, the standard normal distribution is plotted for comparison.

we have investigated such an error by performing a statistical analysis of 100 measurements for the optimal set of angles. This provided us with a way to estimate the error bars on the quantum circuit output in terms of the population states. In detail, we assumed the deviation from the symmetric case, i. e., $P_{|00\rangle} = P_{|11\rangle}$ and $P_{|01\rangle} = P_{|10\rangle}$, so $\Delta P_{|00\rangle,|11\rangle} = P_{|00\rangle} - P_{|11\rangle}$ and $\Delta P_{|01\rangle,|10\rangle} = P_{|01\rangle} - P_{|10\rangle}$. The detailed analysis for one example output distribution can be found in the Supplementary Material.

Finally, to investigate the role of hardware connectivity in this quantum algorithm, we performed the same experiments on the pair D3 and A6, with A6 (the highest-frequency qubit in the pair) as the target qubit. As discussed in the Supplementary Material, to optimize the performance of the CNOT gate between D3 and A6 when A6 is both the flux-tuned qubit and the target one in the register, a counter-phase Z-gate has been included in the CNOT decomposition to address depolarizing errors due to single-qubit phase-accumulation. The Gaussian distribution, obtained for an optimal angle $\theta_1 = 224°$ (Fig. 6 (b)), differs from that obtained for D3-C4, and is more

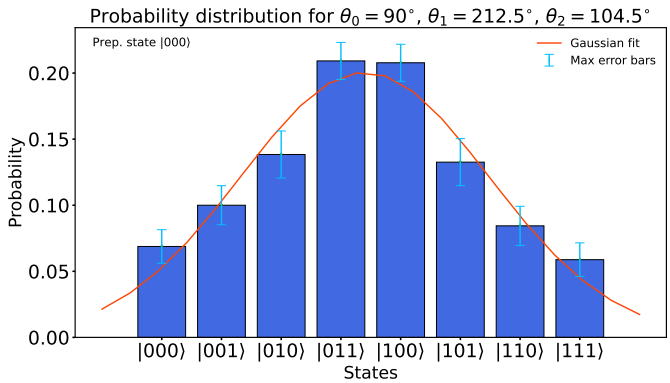


Fig. 7: Three-qubit Gaussian distribution for qubits D3-A6-C4.

consistent with a standard normal distribution (highlighted in green). This result suggests that not only is it fundamental to optimize rotation angles on the quantum machine after prior optimization angles at the classical level, but also that the optimal rotation angles depend on the physical and hardware parameters of the particular pair of qubits considered in the quantum register. This dependence cannot be fully captured by offline hardware-agnostic optimization.

B. Three-qubit normal distribution encoding and loading

For the three-qubit distribution sampling, we used the triplet D3-A6-C4, where the implemented quantum circuit is reported in Fig. 3. First, we fixed $\theta_0 = 90^\circ$ of the $R_y(\cdot)$ gate on D3 and changed θ_1 in the range $[90^\circ, 450^\circ]$ with a step of 36° for the $R_y(\cdot)$ gate on A6, and θ_2 in the range $[90^\circ, 450^\circ]$ for the $R_y(\cdot)$ gate on C4. Again, the goal was to study the output distributions as a function of the angle, also obtaining a transition from positive to negative concavity, passing through a uniform output distribution (see the Supplementary Material). Then, we adjusted the step on the angles in a range for which the output distribution better satisfies the requirements of a gaussian-like output distribution, i.e. $100^\circ < \theta_1 < 250^\circ$ with a step of 7.5°, and $90^\circ < \theta_2 < 380^\circ$ with a step of 14.5°. The optimal angle combination for the D3-A6-C4 register is $\theta_0 = 90^\circ$, $\theta_1 = 212.5^\circ$, and $\theta_2 = 104.5^\circ$ (Fig. 7). As for the two-qubit case, we estimated errors for probability amplitudes by performing a statistic of 100 measurements and by computing the differences of amplitudes between symmetric states.

The experimental evidence of our ability to fine-tune the rotation angles for up to three qubits at a hardware level to achieve a specific output distribution demonstrates that the hardware and the experimental variational approach used here scale well with the number of qubits. We can then legitimately address a more compelling quantum circuit, such as the GCI quantum circuit.

C. GCI quantum circuit

Our hardware-tested GCI circuit instantiates the one-risk-factor case: a latent $Z \sim \mathcal{N}(0,1)$ perturbs the asset's default probability around $p_0 = 0.25$ with sensitivity $\rho = 0.027$. The algorithm requires three qubits and comprises a 2-qubit Gaussian sampling algorithm and one asset preparation and rotation gate, represented by a third qubit (Fig. 1). This quantum circuit includes a controlled-Y (CY gate) that is prevented by the connectivity of our hardware. Therefore, we first transpiled the circuit using Qiskit's preset pass manager, configured for hardware-aware optimization on the target backend. This pipeline performs comprehensive algebraic and structural simplifications (e.g., commutation, cancellation, and resynthesis) while respecting the backend's basis gates and coupling constraints. For both initial qubit assignment and subsequent routing, we employed the SABRE (SWAP-Based Bidirectional heuristic search algorithm) [33], which prioritizes mappings that reduce SWAP insertion and overall circuit depth on constrained topologies. Gate translation utilized synthesis-based methods with an approximation budget to allow slightly inexact unitary realizations in exchange for fewer two-qubit gates and shorter depths —an advantageous trade-off for NISQ hardware where fidelity is limited by circuit length. The final GCI transpiled quantum circuit is reported in Fig. 8.

Initially, the implementation of this quantum circuit on the machine led to unsatisfactory results, with significant deviations from the simulated output. Remarkably, by systematically investigating the output of each quantum subcircuit's layer through both the Qiskit simulator and the actual hardware, the output of the subcircuit in the blue box of Fig. 8 results in a uniform three-qubit distribution. The investigation detailed in Sec. IV-B confirmed that it is possible to experimentally recover a uniform distribution by exploiting the quantum circuit in Fig. 3 and setting specific rotation angles (see Supplementary Material). Therefore, we have opted for replacing the SABRE-based transpiled sub-circuit in the blue box with the one in Fig. 3. Specifically, we exploited qubits D3 and A6 to generate the Gaussian distribution (green square in Fig. S15), representing the financial variable z related to the Probability of Default. We recall that the optimal combination of rotation angles that gives a standard normal distribution for this pair is $\theta_0 = 90^\circ$ on D3 and $\theta_1 = 224^\circ$ on A6. The asset is represented instead by C4. The remaining gates then account for the transpiled realization, directly at the hardware level, of the asset preparation and rotation gates, conditioned by the target normal distribution. In other words, $R_Y(\theta_2)$ and $R_Y(\theta_3)$ on $C4$, and $R_Y(\theta_4)$ on $D3$ are initially derived from the offset and slope of the linearized probability of default in the noiseless quantum circuit model by the Adam and SABRE optimizers. After transpilation and optimization, the circuit no longer preserves a direct correspondence between these angles and the original offset/slope parameters. Therefore, in the final hardware-ready transpiled quantum circuit, they should be regarded as tunable hyperparameters whose values are specific to the instance considered here. Hence, we propose here to use the pulse-level variational approach used for the optimization of the Gaussian distribution, for the search of the optimal rotation angles in the final hardware-ready transpilation of the GCI quantum circuit.

We performed experiments by changing $\theta_{3,4}$ angles in the interval $[0^\circ, 360^\circ]$ to adapt the algorithm efficiency to the hardware, while keeping the asset angle rotation 90° fixed

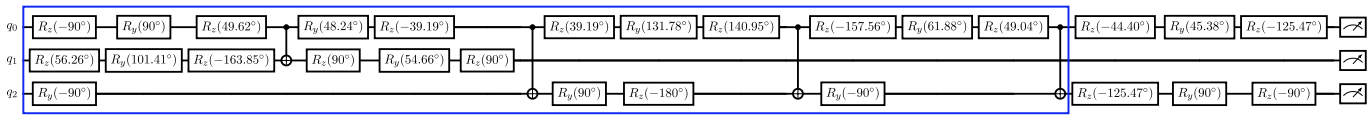
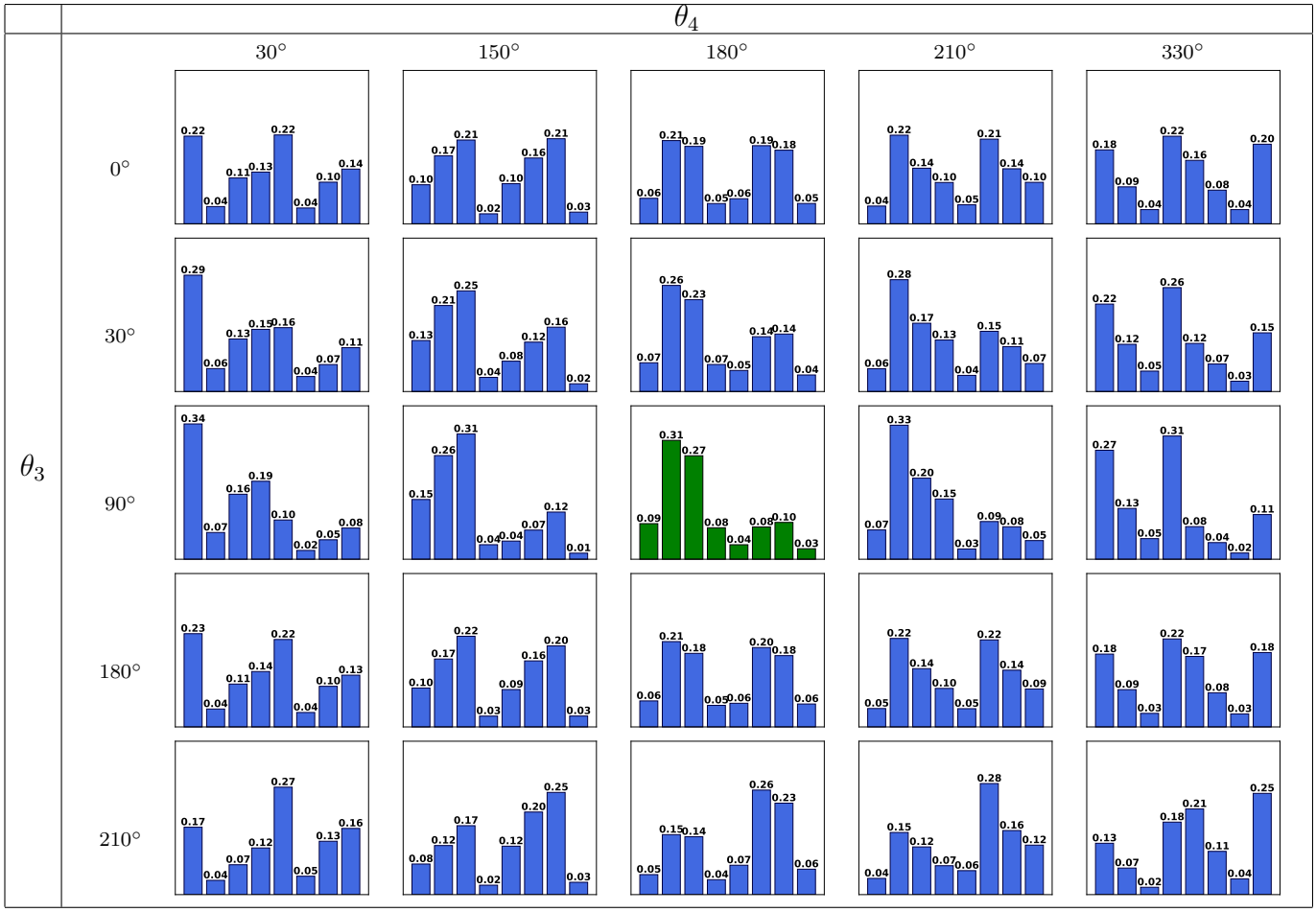


Fig. 8: Use-case GCI transpiled quantum circuit. In the blue square, the sub-circuit provides an output equivalent to a three-qubit uniform distribution circuit.

TABLE III: GCI circuit outputs as a function of θ_3 and θ_4 at fixed $\theta_2 = 90°$. In green, the use-case circuit output. In each plot is depicted the Probability Amplitude for each state in $\{|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle\}$ basis of D3-A6-C4 three-qubit register.



$R_Y(\theta_2)$ (red square in Fig. S15). In Tab. III, we collect the experimental output of the GCI hardware-ready transpiled quantum circuit in Fig. S15 as a function of the $\theta_{3,4}$ rotation angles. $R_Y(\theta_3)$ (purple square in Fig. S15) affects the height of the left or right gaussian with a periodicity of $90°$, while $R_Y(\theta_4)$ (blue square in Fig. S15) acts on the single gaussian symmetry, with optimal behavior obtained for $R_Y(\theta_4) \in [150°, 210°]$. The profile that is most similar to the simulated output of the quantum circuit transpiled using Qiskit's preset pass manager (Fig. 10 (a)) is highlighted in green and obtained for $\theta_3 = 90°$ and $\theta_4 = 180°$. As for the asset angle $R_Y(\theta_2)$ (red square in Fig. S15), in Tab. 2 of the Supplementary Material, we show that it is the experimental parameter that controls the symmetry between the two Gaussians at optimal $\{\theta_3, \theta_4\} = \{90°, 180°\}$.

Finally, we have used these data to calculate the CDF as follows. The pre-processing stage decomposes each measured bit string into two logical components: the leftmost bits encode default indicators, while the rightmost bits represent the latent variable Z. For every observed outcome with an associated probability, the rightmost bits are converted to an integer index, allowing the marginal probability $P(Z = z)$ to be accumulated. The default bits determine which debtors have defaulted; whenever a default bit is equal to one, the corresponding probability contributes to the marginal default probability, and the LGD value is added to the total loss for that scenario. After iterating over all outcomes, the procedure constructs (i) arrays of scenario losses and (ii) their probabilities, from which the expected loss is computed. Identical loss values are collapsed to form a discrete probability density function (e.g., the PDF) and its cumulative distribution function (e.g., the CDF). In Fig. 10, we compare the CDF for the simulated

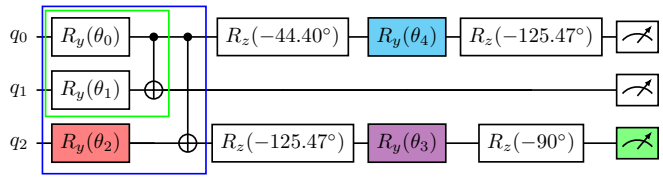


Fig. 9: Transpiled Gaussian Conditional Independence model quantum circuit. The first CNOT is here decomposed in terms of Hadamard and CZ gates, and includes a counter-phase Z gate with an angle of $-135°$ to counteract CZ depolarizing errors, as discussed in Supplementary Material. The green box represents the 2-qubit normal distribution loading, while the blue box replaces the blue one in Fig. 8.

transpiled quantum circuit with Qiskit's preset pass manager (Fig. 8), the simulated hardware-ready transpiled quantum circuit (Fig. S15), and the experimental implementation on the hardware of the latter. The circuits' outputs are reported in panels (a), (b), and (c), respectively.

The CDF curves calculated from the experimental data (affected by an error of $\sim 5\%$) obtained with the hardware-ready transpilation on the machine and the simulated one obtained with the initial transpilation, respectively, place the 95% VaR at the same total-loss level ($L = 1000$), and the plotted CDF values remarkably coincide at the two observed points: $P(L \leq 0) \approx 0.75$ and $P(L \leq 1000) = 1.0$. Within numerical tolerance and the experimental errors, the quantum and classical outputs are indistinguishable on this grid—yielding the same VaR, the same CDF shape between 0 and 1000, and the same risk conclusion for this scenario. Therefore, this confirms the validity of our experimental approach, which is entirely based on the performances of the NISQ hardware at hand. Indeed, although the noiseless simulation of the hardware-ready transpiled quantum circuit used for the experiment leads to an output distribution fairly consistent with the experimental output, the corresponding 95% VaR is placed at a slightly different loss level, indicating that the simulator cannot replicate the experimental outcome since it does not include a priori any information about the noisy behavior of the hardware, which is instead implicitly taken into account by the variational hyper-tuning of the hardware-ready transpiled circuit on the machine. This result confirms that, although transpilation should only effectively re-assign and re-combine unitary rotations in a quantum circuit to address connectivity and circuit depth, it does not consider any knowledge about the sensitivity of an NISQ device to specific noise sources, which is instead a fundamental requirement in the NISQ era.

V. Conclusion

In this work, we have provided a systematic experimental investigation of how variational quantum circuits can model distributions relevant to CRA on a superconducting quantum computer: two- and three-qubit Gaussian distributions and the GCI.

We have examined how sQPU connectivity, the different circuit nature of superconducting qubits in quantum registers, and the quantum circuit transpilation itself (e.g., entangler placement and single-qubit rotation settings) influence the desired distribution, encoded in the probability amplitudes of the prepared quantum states. A remarkable result is that the rotation angles required to realize a normal-like histogram are not generic but depend on the specific hardware characteristics of the device. We have experimentally analyzed the impact of single-qubit rotation angles on the shape of probability amplitudes distributions encoded in both two- and three-qubits registers. This allowed us not only to identify the optimal angle configuration but also, and most importantly, to determine the experimental confidence error threshold in a typical NISQ device by implementing hundreds of experimental runs. We specify here that no readout error mitigation technique has been applied, nor have we integrated quantum-noise limited cryogenic amplifiers at the hardware stage. Although we envision including both into our quantum algorithms implementation pipeline in the near term [29], [34], [35], our investigation demonstrates that the readout fidelities achieved on the processor already offer satisfying results. The in-situ variational approach introduced here has also been used to optimize the GCI quantum circuit output for fixed problem parameters, taken here as an example, obtaining excellent agreement with the classical counterpart. In a sense, we have performed a hardware-aware transpilation of the quantum circuits directly at the pulse level for this use case, allowing us to capture the realistic nature of the hardware, down to the level of the individual qubits' performances and connectivities. The validity of our approach has been confirmed by the calculation of the CDF for a specific use case, obtained by using the experimental data derived from an *ad-hoc* transpilation of the quantum circuit, implicitly taking into account the errors of the machine, and by simulating the same quantum circuit with a noiseless simulator. Such dependence is critical in the NISQ era and emphasizes the requirement for a clear understanding of quantum computing in all its parts to achieve quantum utility in the near term: from algorithm coding to implementation through analog pulses, which allowed us to construct a more efficient quantum circuit in terms of gate depth.

We envision that more and more research centers and industries will rely on proprietary superconducting hardware designs in the near term, with their own geometries, connectivity, characteristic circuits, and electronic parameters for quantum gate implementation. We believe this work provides a guide towards the hardware-aware implementation of other proof-of-concept quantum algorithms and motivates the use of quantum compilers that adapt circuit structure and parameterization to the specific QPU, rather than relying on a fixed, hardware-agnostic configuration.

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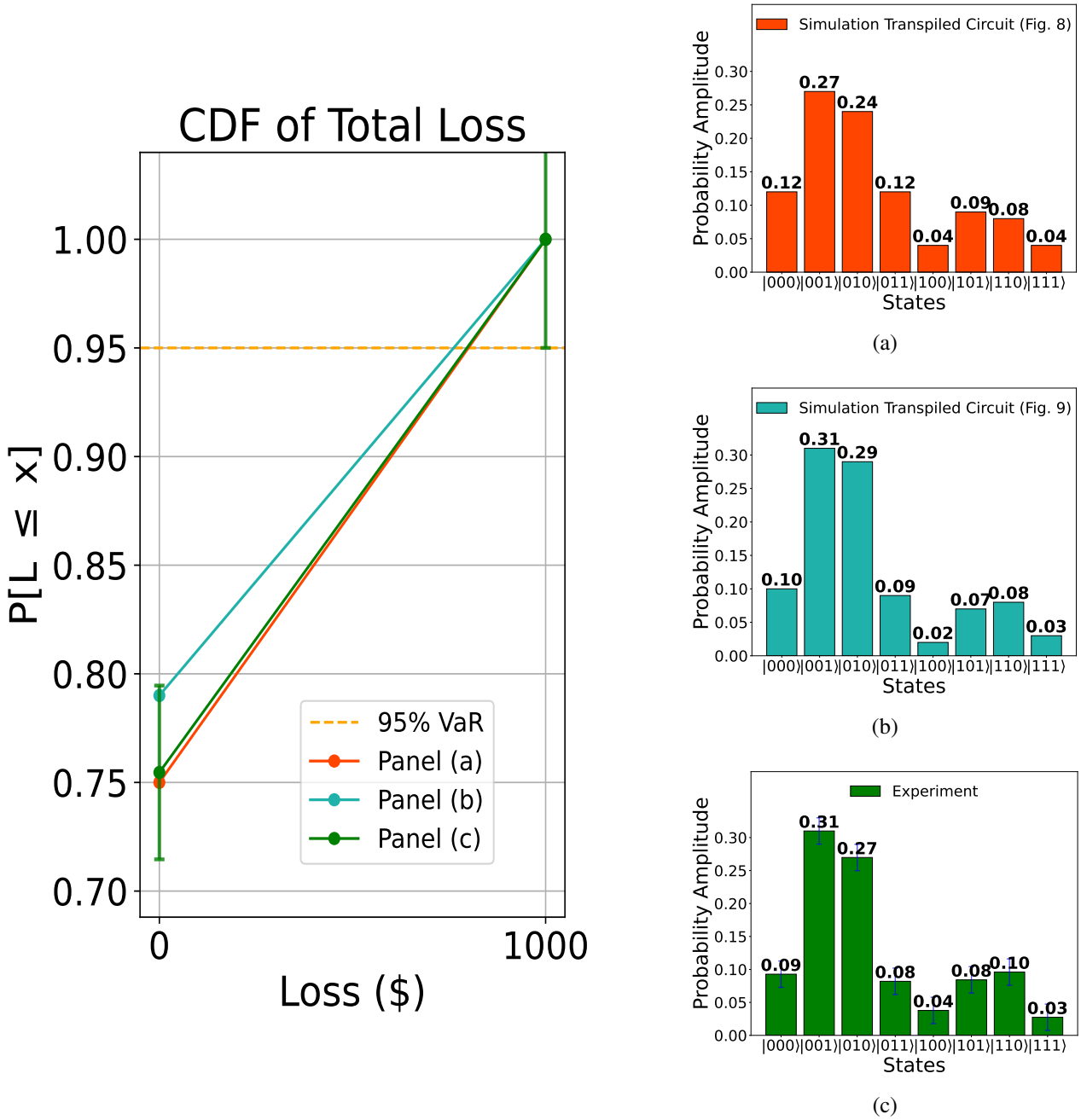

(a)

(b)

(c)

Fig. 10: Cumulative Distribution Function (CDF) of total losses for a loss $L = 1000\$$ and a VaR of 95%, calculated from the probability distribution output of the GCI model in (a) (simulation of the transpiled quantum circuit in Fig. 8), in (b) (simulation of the transpiled quantum circuit in Fig. S15), and in (c) (experimental output of the same transpiled circuit).

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

VI. Conflict of Interest Statement

The authors declare no conflicts of interest.

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Supplementary Materials for Quantum Circuit-Based Adaptation for Credit Risk Analysis

VII. Conditional default probability as quantum state probability: a mathematical demonstration

A. Gaussian Conditional Independence Model

In the Gaussian conditional-independence model [1], [2], the default probability at the obligor-level conditioned on a latent Gaussian factor z is

$$PD_k(z) = \Phi\left(\frac{\Phi^{-1}(p_k^0) - \sqrt{\rho_k}\, z}{\sqrt{1-\rho_k}}\right), \tag{17}$$

with the default baseline level p_k^0, the correlation of assets ρ_k and Φ the standard normal Cumulative Distribution Function (CDF). Our circuit is designed so that the probability of measuring $|1\rangle$ on a target qubit matches this conditional default probability. We write

$$PD_k(z) \equiv \Pr(|1\rangle) = \sin^2\big(\alpha_k z + \beta_k\big), \tag{18}$$

which is equivalent to linearizing

$$g(z) := \arcsin\sqrt{PD_k(z)} \approx \alpha_k z + \beta_k \quad \text{around } z = 0. \tag{19}$$

Let f denote standard normal PDF, and define

$$\psi_k := \frac{\Phi^{-1}(p_k^0)}{\sqrt{1-\rho_k}}. \tag{20}$$

The offset is the zeroth-order term of g:

$$\beta_k = g(0) = \arcsin\sqrt{\Phi(\psi_k)}. \tag{21}$$

By the chain rule, the slope is

$$\alpha_k = \left.\frac{dg}{dz}\right|_{z=0} = -\frac{1}{\sqrt{1-\Phi(\psi_k)}}\cdot\frac{1}{2\sqrt{\Phi(\psi_k)}}\cdot f(\psi_k)\cdot\frac{\sqrt{\rho_k}}{\sqrt{1-\rho_k}}. \tag{22}$$

Hence the first-order approximation reads

$$\arcsin\sqrt{PD_k(z)} \;\approx\; \alpha_k z + \beta_k, \qquad PD_k(z) \;\approx\; \sin^2\big(\alpha_k z + \beta_k\big), \tag{23}$$

so the required single-qubit gate is a linear-rotation block

$$R_y\Big(2(\alpha_k z + \beta_k)\Big). \tag{24}$$

When the latent factor is represented on an n-qubit register discretizing $[-z_{\max}, z_{\max}]$ with grid step

$$\Delta z = \tfrac{2 z_{\max}}{2^n - 1}, \tag{25}$$

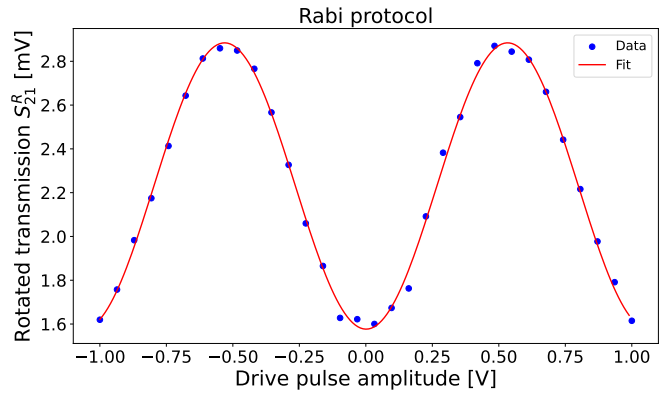
the circuit angles are rescaled to the coded grid:

$$\tilde{\alpha}_k = \Delta z\, \alpha_k = \frac{2 z_{\max}}{2^n - 1}\,\alpha_k, \qquad \tilde{\beta}_k = \beta_k - \alpha_k z_{\max}, \tag{26}$$

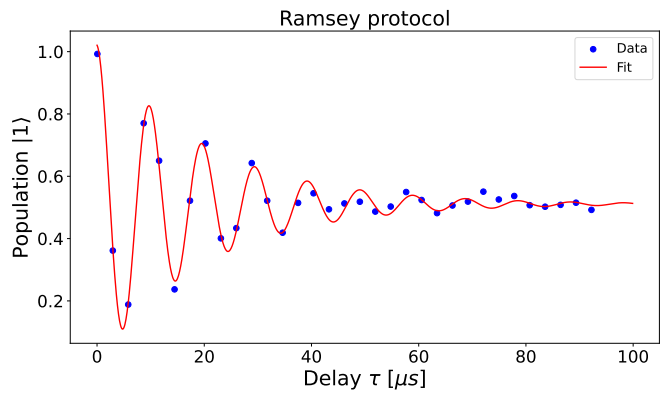
and the implemented gate becomes

$$R_y\Big(2\big(\tilde{\alpha}_k\, \hat{z}_{\text{code}} + \tilde{\beta}_k\big)\Big), \tag{27}$$

where $\hat{z}_{\text{code}} \in \{0, 1, \ldots, 2^n - 1\}$ denotes the integer basis value encoded by the n-qubit z-register.



(a)



(b)

Fig. S1: In **(a)**: example of a Rabi oscillation measurement; in **(b)**: example of a Ramsey Interferometry experiment.

VIII. Calibration, Fidelity and Readout

In this section, we describe the main protocols used for calibration and optimization of single-qubit X and Y gates, how to benchmark single-qubit gate fidelities and report on the acquired experimental data obtained before running the experiments discussed in the main manuscript.

A. Drive signal calibration protocols

The first step of the drive signals' calibration process consists in time-domain measurements. The Rabi oscillations measurements provide the estimation of the π-pulse amplitude, i. e. the microwave drive signal amplitude necessary to bring the qubit from the ground state $|0\rangle$ to the first excited state $|1\rangle$ for a fixed drive pulse duration. The operative procedure to measure the Rabi oscillations consists in sending a drive signal followed by a readout pulse, by changing the amplitude of the drive tone.

Ramsey interferometry protocol allows to tune the drive signal frequency by preparing the qubit on the Bloch sphere equator applying a $X_{\frac{\pi}{2}}$-pulse (i. e. halving the π-pulse amplitude measured throuh Rabi oscillations) and then leaving the qubit free to evolve spontaneously for a time τ before that a second $X_{\frac{\pi}{2}}$-pulse projects the Bloch vector back to the z-axis. When inferring the qubit state, in presence of

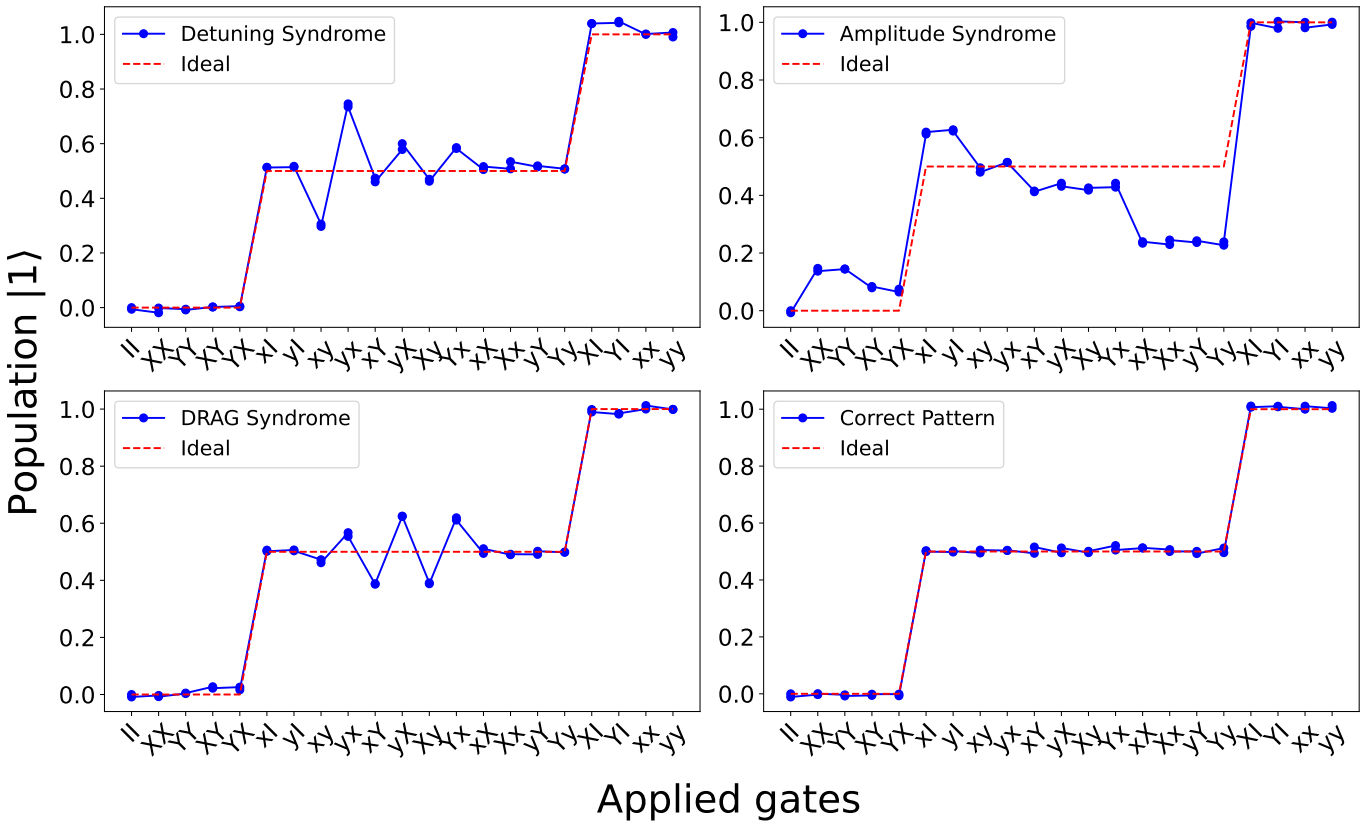


Fig. S2: Pattern signatures for of power amplitude, detuning and DRAG syndromes, and corrected pattern, showing the ideal z projection as a function of *AllXY* sequences.

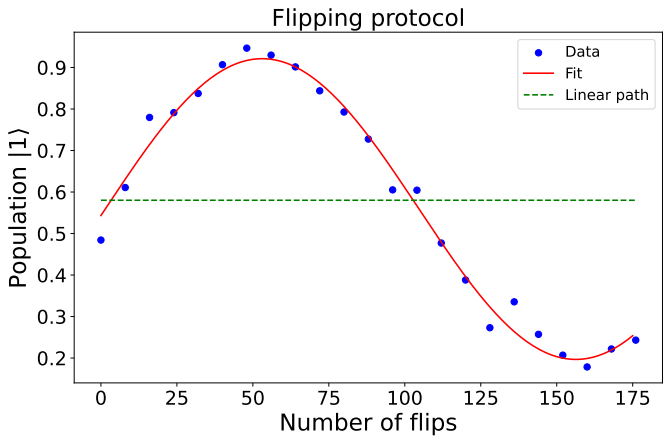
detuning between the drive and qubit frequencies, i.e. when the drive signal is off-resonance with the qubit frequency, damped oscillations with an exponential decay arise. On the contrary, the output features only an exponential decay.

B. Detecting syndromes for control optimization: All-XY sequence

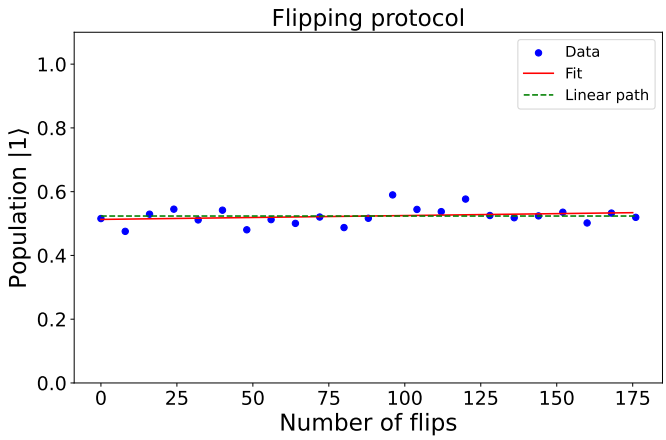
Higher quality rotations are desirable for applications where a high fidelity is needed, as for quantum algorithms implementation. The *AllXY* sequence tests the result of all combinations of one or two single-qubit gates of the X and Y type being sensitive to a variety of error syndromes [3]. During the protocol, 21 different pulse combinations, belonging to the "Clifford set" of quantum gates [4], are performed on a qubit, producing rotations around the x- and y-axes by an angle of $\pi/2$ or π before measuring its state. Each pulse combination is sensitive to various errors, which will have a distinct fingerprint in the deviation from the ideal response (Fig. S2), providing a quick way of diagnosing syndromes [3]. It is possible to detect mainly three types of syndromes, which can be tackled with a specific correction protocol:

- Power syndrome: related to drive signal amplitude, it yelds a characteristic "step" pattern, shown in Fig. S2, due to over- or under-rotations of the gates. To correct amplitude errors, in this work we used the **Flipping** (or error amplification) protocol. This protocol consists in preparing the qubit on the equator of the Bloch sphere by applying a $\pi/2$-pulse. Then, $2N$ π-pulses are applied, with an increasing N (Fig. S3 (a)). Each additional π-pulse amplifies the power error syndrome, causing the qubit state to oscillate as a function of N. By fitting these oscillations, it is possible to identify the control pulse amplitude to correct this behavior, and observe instead a linear dependence of the qubit state population (Fig. S3 (b)).
- Detuning syndrome: caused by the detuning between drive signal frequency and qubit resonance frequency, it yelds a characteristic "zig-zag" pattern on the x and y rotations. Therefore, this syndrome can be corrected by implementing Ramsey protocol to fit the detuning and estimate the correction needed to set right drive signal frequency.
- DRAG syndrome: related to phase errors, it's mainly due the IQ mixer imperfections, such as amplitude imbalance and skewness. The DRAG technique corrects both gate phase errors and leakage caused by the presence of higher excited states of weakly anharmonic transmon qubit [5]. Through **Motzoi** protocol, it is possible to efficiently tune-up the *DRAG* parameter [6] as follows: by taking two of the *AllXY* pulses which exhibit the opposite sign of error (for instance $Y_\pi X_{\frac{\pi}{2}}$ and $X_\pi Y_{\frac{\pi}{2}}$) and performing them as a function of the parameter proportional to the *DRAG* derivative correction, one obtains two lines that cross at the point where the *DRAG* optimal coefficient is

optimal (Fig. S4).



(a)



(b)

Fig. S3: Flipping protocol before **(a)** and after **(b)** applying power error syndrome correction.

C. Gate fidelity

In the main manuscript, we used the Randomized Benchmarking (RB) and the Interleaved Randomized Benchmarking (IRB) protocol in order to assess the average single-qubit gate fidelity and the CZ two-qubit gate fidelity, respectively [7].

The Randomized Benchmarking protocol performs an average gate fidelity evaluation that is robust to State Preparation and Measurements (SPAM) errors, gaining information on the hardware gate implementation [8]. RB protocol consists of applying a random sequence of N gates, randomly picked from the Clifford gate set $\{\mathcal{C}_i\}$, followed by their inverse Clifford gate before measuring the qubit state population. Ideally, the measured state should be identical to the initial state, i.e. ground state $|0\rangle$, but the measured state will differ due to the errors induced by the gate implementation. The number of gates in a sequence is chosen to be $N_{Cl} = [1, 2, 4, 8, 16, 32, 64, 128, 256, 512, ...]$ to easily obtain an exponential decay trend (Fig. S5). To perform repeated measurements with different gate combinations, for each sequence N_{Cl}, random gates are sorted through a certain

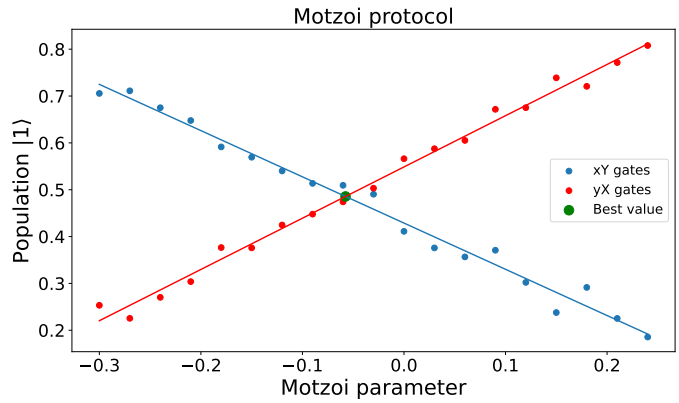


Fig. S4: Motzoi protocol is performed by choosing two of the *AllXY* pulse sequences with opposite sign errors, and measure them both with varying *DRAG* coefficient: the optimal DRAG value is located at the crossing point of two resulting lines [3].

number of seeds, i.e. random number generators. Therefore, the survival probability $F_{|0\rangle}$, i.e. the average sequence gate fidelity is measured for each seed. After the measurement, the depolarizing error p is fitted using the model

$$F_{|0\rangle} = Ap^{N_{Cl}} + B. \tag{28}$$

Here A and B parameters give information about SPAM errors. Then, the gate fidelity over all the Clifford sequences is given by [8]:

$$F_{Cl} = 1 - \frac{d-1}{d}(1-p). \tag{29}$$

Lastly, it is possible to get an estimation of average gate fidelity per applied gate as [8]:

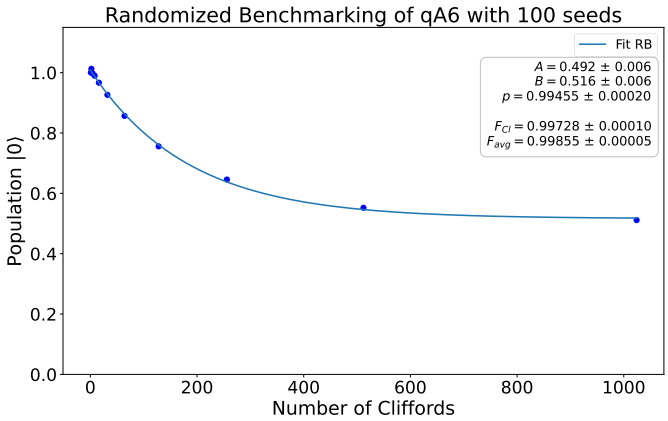
$$F_{avg.gate} = (F_{Cl})^{\frac{1}{1.875}}, \tag{30}$$

where the exponent of the root-square is related to the decomposition of each gate into the standard minimal sequence of π- and $\pi/2$-pulses around the x and y axes, requiring an average $\langle N_P \rangle = 1.875$ pulses per Clifford. In Fig.S5, we show the RB experiment performed simultaneously on the three qubits used for the implementation of the quantum circuits in the main manuscript.

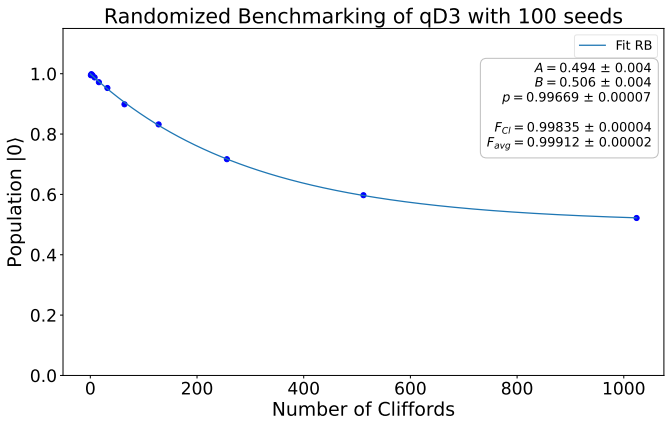
While the RB allows for estimating the average single-qubit gate fidelity, the Interleaved Randomized Benchmarking (IRB) allows for estimating the average error of an individual quantum gate. Therefore, the CZ gate (the native two-qubit gate in our processor) fidelity can be estimated by interleaving the CZ gate into the standard RB protocol [9]. First, a standard RB experiment on the two-qubit allows to recover the depolarizing error p_0 without the insertion of the CZ gate in the sequence. Then, the RB is repeated by including the CZ after each Clifford gate randomly applied. As for the standard RB, the depolarizing error p_1 can be fitted by the exponential decay. Finally, the average fidelity for the CZ gate can be estimated by:

$$F_{cz,naive} = (F_{Cl})^{\frac{1}{1.5}}. \tag{31}$$

In Fig.S6, we show the IRB experiment for the two pairs investigated in the main manuscript.



(a)



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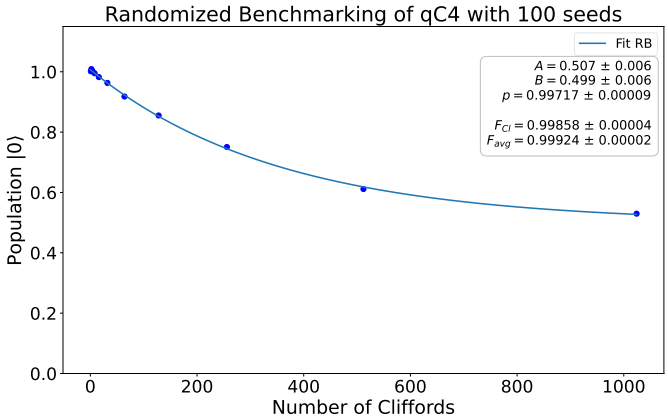
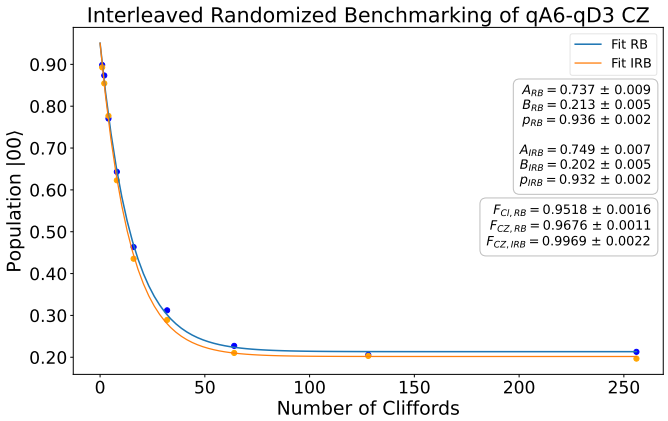
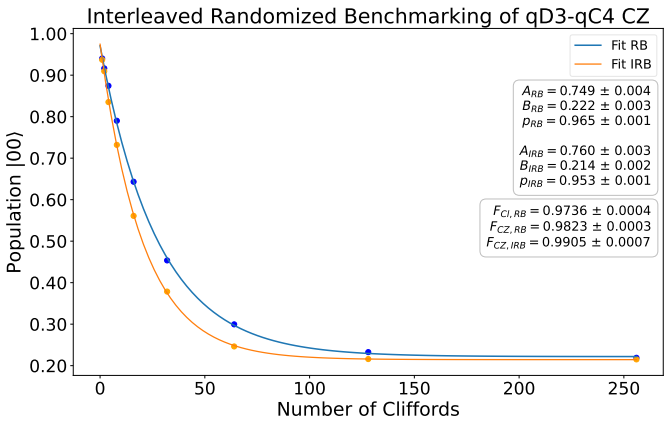


(c)

Fig. S5: Simultaneous Randomized Benchmarking protocol with 100 seeds of qubits A6, D3, C4.



(a)



(b)

Fig. S6: Interleaved Randomized Benchmarking protocol for CZ gate implemented on qubit couples A6-D3, and D3-C4.

D. Readout

A fundamental requirement for the implementation of quantum algorithms is the ability to readout the state of a quantum register in such a way to prevent the quantum state collapse, i. e. within a quantum non-demolitive readout. Transmon qubits allows to do it by dispersively coupling each qubit with a superconducting readout resonator. As detailed in Ref. [12], the assignment state probability of a qubit is obtained by measuring multiple times the signal across the resonator (Single-Shot Readout - SSRO) and counting how many times the qubit has been in the computational basis state $\{|0\rangle, |1\rangle\}$. This eventually requires to identify a discrimination threshold, which is obtained through an initialization experiment: the qubit is prepared in the two computational basis states, SSRO is performed and the discrimination threshold is identified. The procedure can be extended to more qubits, and the counting mechanism used in this work exploits the conditional readout proposed in Ref. [12]. As a consequence, the ability to efficiently readout the output of a quantum circuit strongly depends on the readout parameters that allows in turn to efficiently discriminate the state. Among them, the microwave excitation tone sent to the resonator, as well as the acquisition signal used to measure the complex voltage across the resonator, in which the qubit state is encoded, have been calibrated by performing repeated SSRO initialization experiments as a function of their pulse parameters (duration, amplitude, frequency): the best parameters have been chosen as the ones for which both leakage towards non-prepared states and readout fidelity was maximized.

Finally, before every quantum circuit we implement the SSRO initialization experiment to identify the discrimination threshold right before any quantum circuit experiment. In Fig.S7, for example, we show the initialization matrix for the

GCI algorithm experiment.

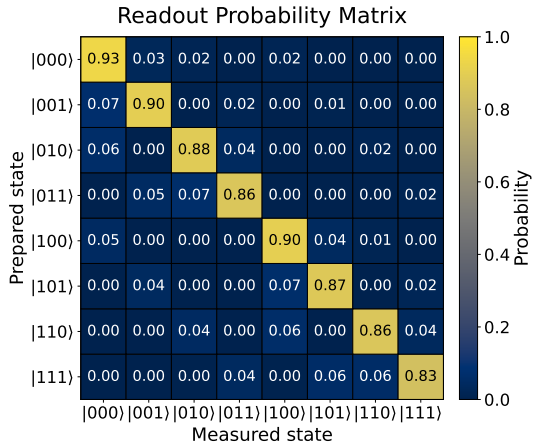


Fig. S7: Single Shot Readout Probability Matrix for the three qubit register D3-A6-C4.

IX. Details on the hardware-aware variational quantum algorithms

A. Two-qubit Gaussian Loading: CNOT implementation

As shown in Section C. of the main manuscript, we investigated the role of the hardware connectivity in the implementation of the quantum algorithm by performing two-qubit Gaussian distribution generation on two different two-qubit registers.

For A6 and D3, we found an optimal combination angle $\theta_0 = 90°$ and $\theta_1 = 222°$. We also tested the CNOT directionality effect by changing the role of control and target qubit. We recall that the $CNOT$ gate is implemented as $CNOT_{0,1} = (\mathbb{I} \otimes H)CZ(\mathbb{I} \otimes H)$ (Fig. S8). Specifically, the implementation of the CZ gate consists of moving the highest-frequency qubit (i. e. the tuned qubit) from the sweet-spot towards the lowest-frequency qubit (spectator qubit). Here, for example, we used a Sudden-Net-Zero Flux pulse, opportunely calibrated. When the tuned qubit corresponds to the control qubit, it will only relive a flux-pulse at the hardware level, since Hadamard gate will be applied on the spectator qubit. If, instead, it is the target qubit, the CZ gate will follow a Hadamard gate. However, although the CZ gate should only

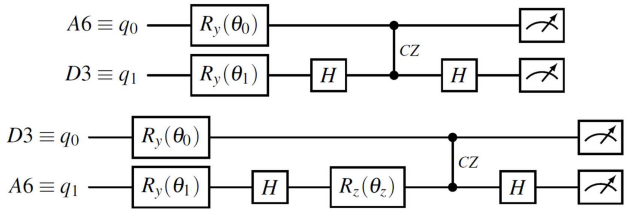


Fig. S8: Two-qubit quantum circuit for the Gaussian loading on A6-D3 register: (top), the implementation of $CNOT_{A6,D3}$, (bottom) the implementation of $CNOT_{D3,A6}$ with counter phase $R_z(\theta)$ gate, i.e. when the tuned qubit is the targeted one.

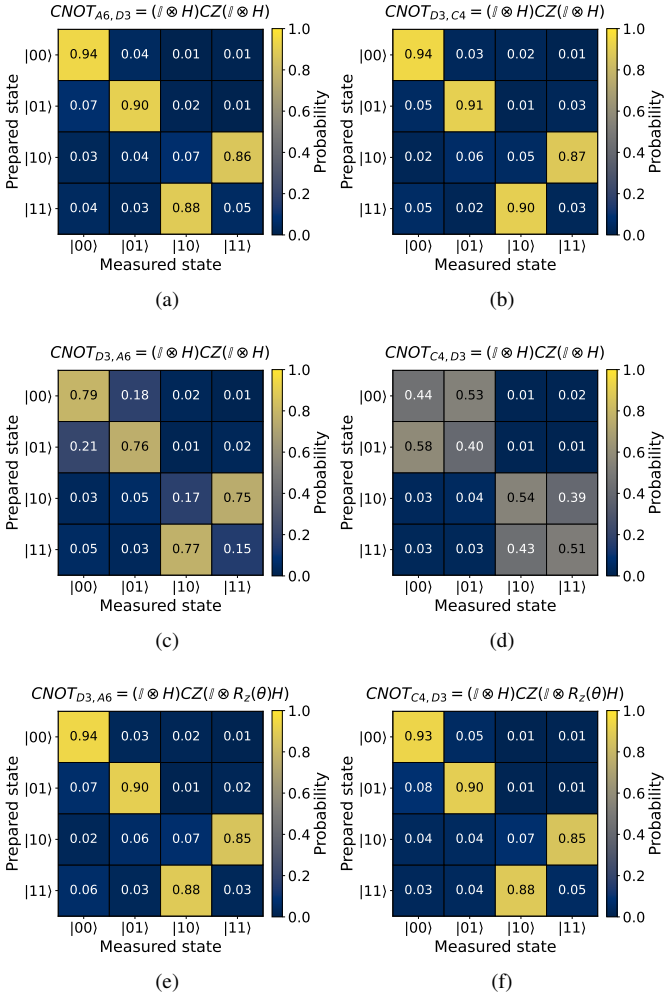
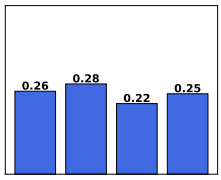
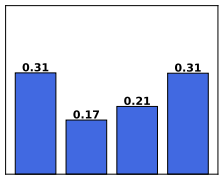
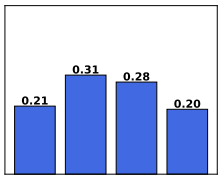
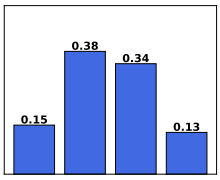
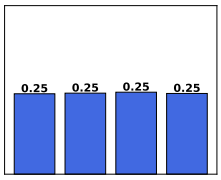


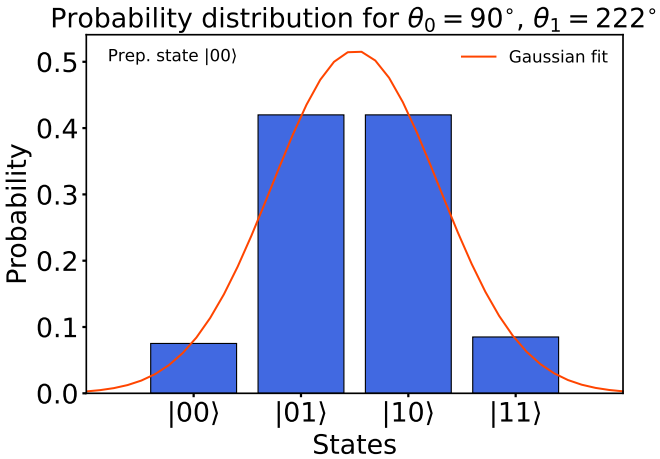
Fig. S9: In **(a)** and **(b)** $CNOT_{A6,D3}$ and $CNOT_{D3,C4}$, respectively. In **(c)** and **(d)** $CNOT_{D3,A6}$ and $CNOT_{C4,D3}$, respectively, i.e. when the tuned qubit is the target of the $CNOT$ gate. In **(e)** and **(f)** $CNOT_{D3,A6}$ and $CNOT_{C4,D3}$ with the counter gate $Rz(\theta)$, with $\theta = -135°$ for D3-A6 register and $\theta = -90°$ for C4-D3 register.

adiabatically move the qubit from the sweetspot, in reality this dynamics may cause the qubit to gain an additional single-qubit phase, compromising the quality of the CNOT. In order to counteract this effect, an additional $R_z(\theta_z)$ gate is added between the first Hadamard gate and the CZ, acting as a counter phase gate. By repeatedly measuring the Truth table of the CNOT as a function of the counter-phase gate angle, we have identified $\theta = -135°$ as the best correction angle for D3-A6 couple and $\theta = -90°$ for C4-D3 couple. The results are reported in Fig.S9.

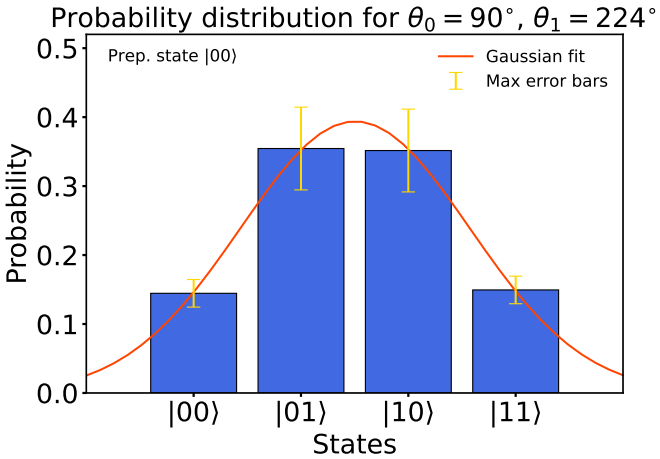
Finally, following this argument, we asked ourselves whether the optimal angles for generating two-qubit Gaussian distributions would depend on the roles of the target and control in the register. Therefore, we performed the algorithm depicted in Fig.S8, changing the role of target and control in the pair. We obtained a similar optimal angle combination, i.e. $\theta_0 = 90°$ and $\theta_1 = 224°$, meaning that the optimal

TABLE S1: Two-qubit circuit output on D3-C4 register as a function of θ_1 and fixed $\theta_0 = 90°$ by preparing the $|11\rangle$ state. For $\theta_1 \in \{90°, 270°\}$ probability amplitudes feature an upside down gaussian-like shape, while for $\theta_1 \in \{270°, 450°\}$ feature a normal gaussian-like shape.

	$\theta_0 = 90°$				
θ_1	90°	111°	279°	300°	447°
					



(a)



(b)

Fig. S10: Two-qubit Gaussian generation for the A6-D3 register: in **(a)** D3 plays the role of target qubit, while in **(b)** A6 is the targeted one.

angles don't depend on the role of the qubits, but only on the exploited two-qubit register (Fig.S10).

B. Alternative state preparation

The goal of the work is to load a two-qubit normal distribution in the GCI quantum circuit. In Tab.S1, we show the same experiment described in Sec. A "Two-qubit normal distribution

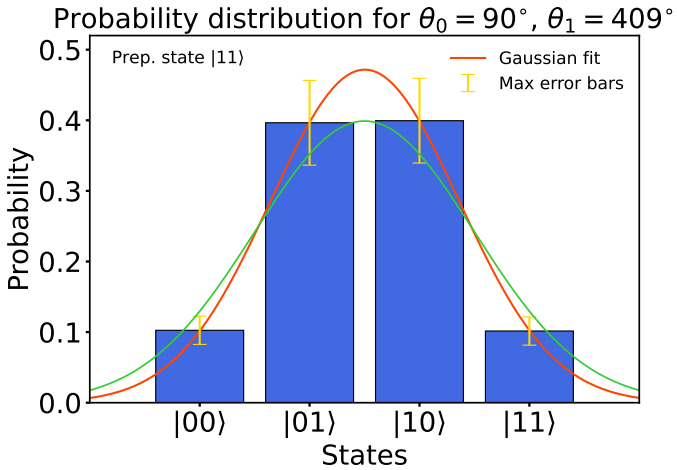


Fig. S11: Two-qubit Gaussian distribution at the optimal θ_0 and θ_1 angle and corresponding Gaussian fit for D3-C4 pair prepared in $|11\rangle$ state. In green the N(0,1) distribution.

encoding and loading", by preparing the D3-C4 qubit pair in $|11\rangle$ initial state state, rather than in the ground state. In this case, the general trend is opposite to what we observe for the initial state $|00\rangle$. We followed the same variational approach discussed in the main text to obtain a standard normal distribution, and in Fig.S11 the gaussian distribution is obtained with the optimal angle configuration $\theta_0 = 90°$ and $\theta_1 = 409°$.

C. Three-qubit Gaussian Loading: variational experiments.

As done for the two-qubit experiment, in Fig. S12 we show the experimental probability amplitudes for the three-qubit basis states as a function of the angle θ_1 and θ_2: they show a reasonable agreement with the theoretical expectation. Remarkably, as discussed in the main text, we can revover a uniform distribution for different angles configurations.

D. Error bars

In order to estimate the error bars on the probability amplitudes, we performed a statistics of 100 measurements for the optimal set of angles. Here we show an example for $\theta_0 = 90°$ and $\theta_1 = 191°$ for the two-qubit register D3-C4. In particular, we assumed the deviation from the symmetric case, i.e. $P_{|00\rangle} = P_{|11\rangle}$ and $P_{|01\rangle} = P_{|10\rangle}$, so

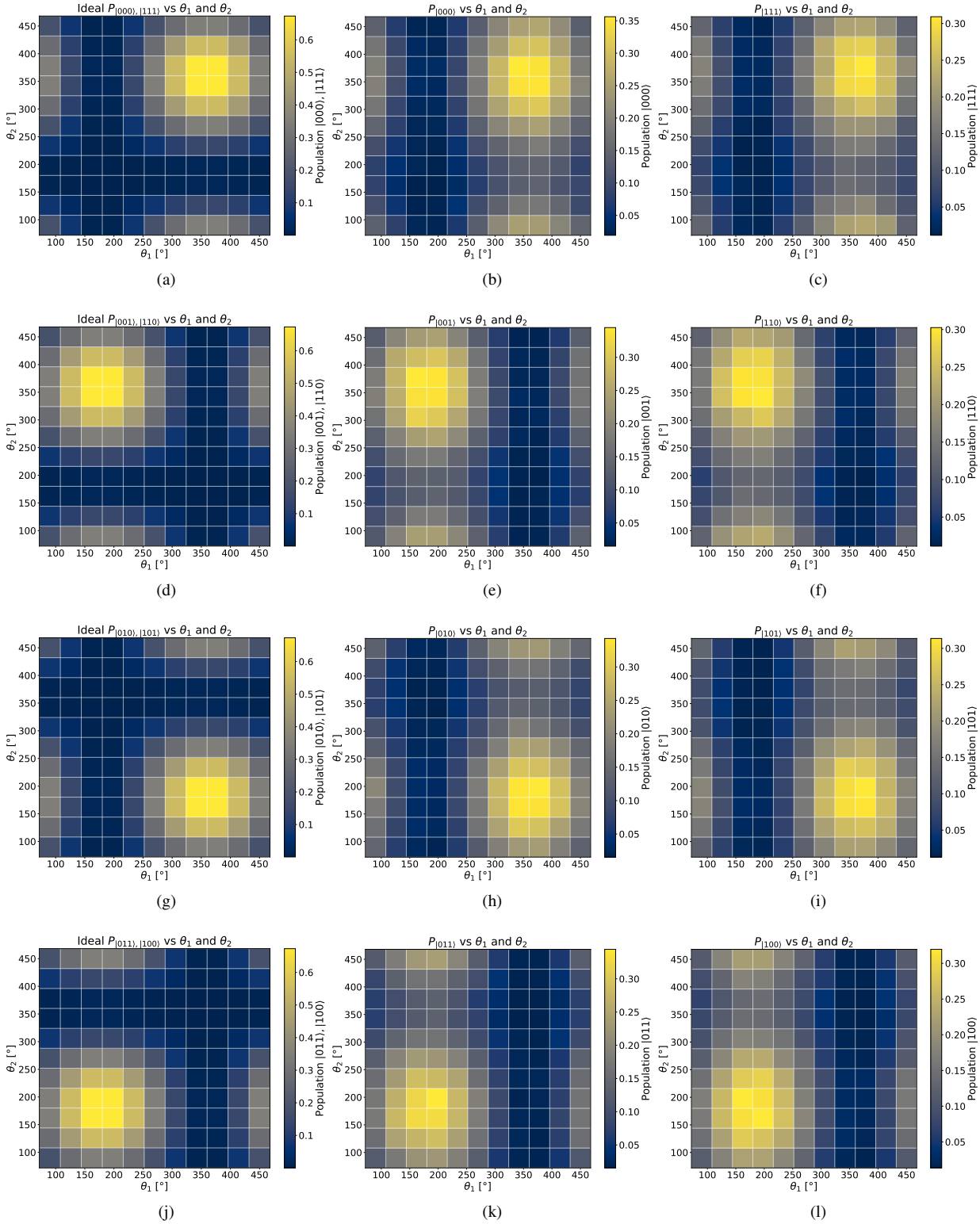
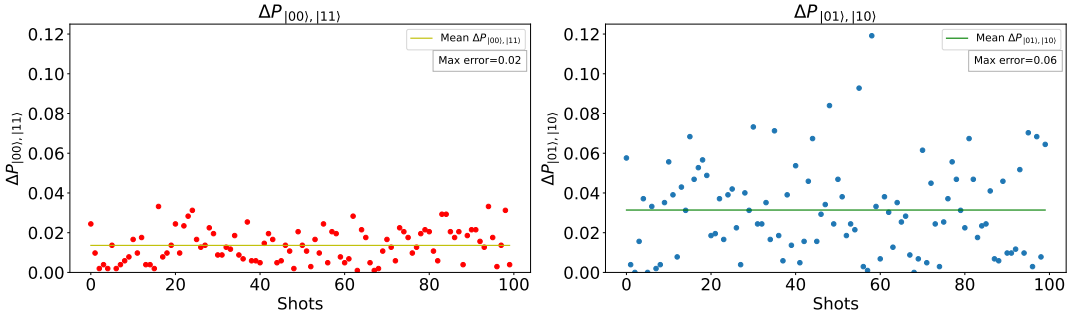


Fig. S12: Comparison between probability amplitudes of $|000\rangle, |001\rangle, |010\rangle, |011\rangle, |100\rangle, |101\rangle, |110\rangle, |111\rangle$ states and theoretical model as a function of θ_1 and θ_2 with fixed $\theta_0 = 90°$

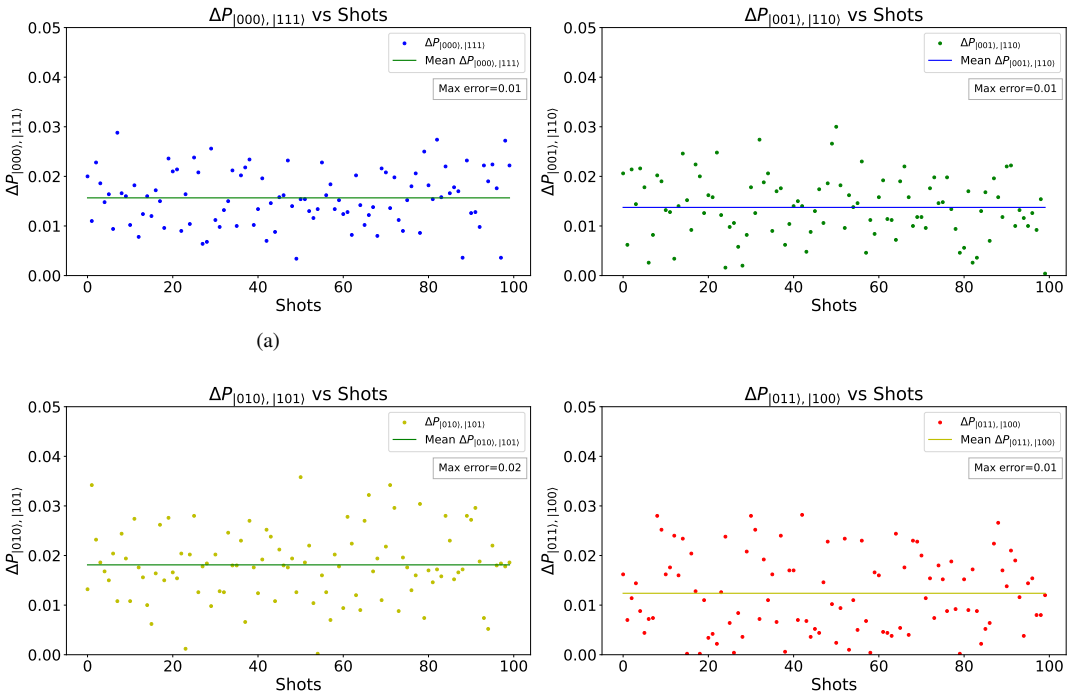
$\Delta P_{|00\rangle,|11\rangle} = P_{|00\rangle} - P_{|11\rangle}$ and $\Delta P_{|01\rangle,|10\rangle} = P_{|01\rangle} - P_{|10\rangle}$. We show in Fig.S13 that the mean values are $\Delta P_{|00\rangle,|11\rangle} = 0.02$ and $\Delta P_{|01\rangle,|10\rangle} = 0.06$.

We proceeded in the same way for the three-qubit gaussian generation experiment by performing a 100 measurements statistic with the optimal angles combination for D3-A6-C4 register: $\theta_0 = 90°$, $\theta_1 = 212.5°$ and $\theta_2 = 104.5°$ (Fig. S14). The mean values are $\Delta P_{|000\rangle,|111\rangle} = 0.01$, $\Delta P_{|001\rangle,|110\rangle} =$



(a) (b)

Fig. S13: Error bars estimation from 100 measurement statistic for the D3-C4 register.



(a) (c) (d)

Fig. S14: Error bars estimation from 100 measurement statistic for the D3-A6-C4 register.

0.01, $\Delta P_{|010\rangle,|101\rangle} = 0.02$ and $\Delta P_{|011\rangle,|100\rangle} = 0.01$.

E. Use-case GCI algorithm: experiments as a function of the asset rotation angle

In Tab.S2, we show the probability amplitudes as a function of $R_y(\theta_2)$, i.e. the asset preparation gate, at fixed $\theta_3 = 90°$ and $\theta_4 = 180°$ (Fig. S15).

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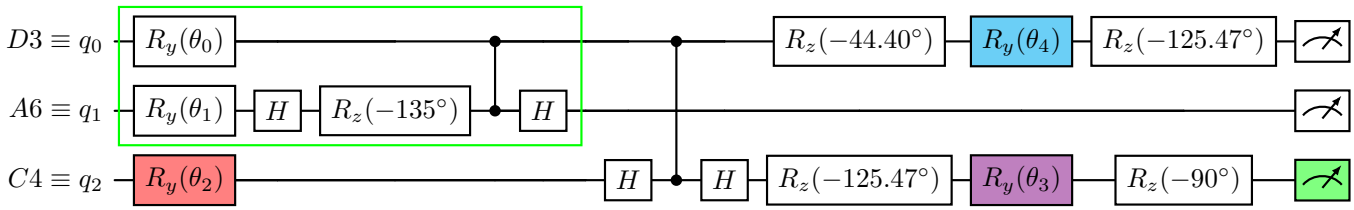
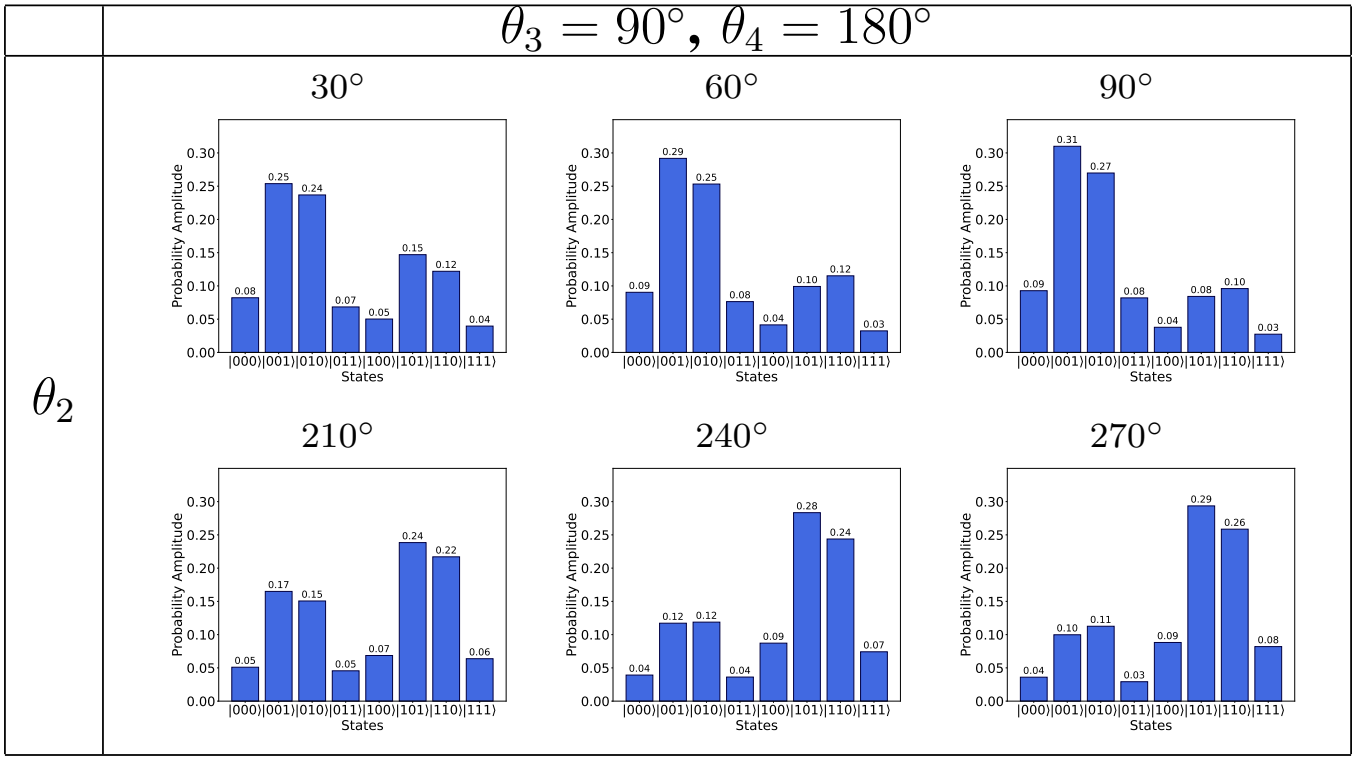


Fig. S15: Use-Case quantum circuit.

TABLE S2: GCI circuit output as a function of rotation asset θ_2 at fixed $\theta_3 = 90°$ and $\theta_4 = 180°$.



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